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#Ashutosh manik malewar 34(A)
from tensorflow.keras import datasets, layers, models
import matplotlib.pyplot as plt
import numpy as np
import tensorflow as tf
(x_train, y_train), (x_test, y_test) = datasets.cifar10.load_dat
```

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Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170498071/170498071 ━━━━━━━━━━━━━━━━━━━━ 327s 2us/step
```

```
x_train = x_train / 255.0
x_test = x_test / 255.0
```

```
import tensorflow as tf
from tensorflow.keras import layers, models
class_names = ['airplane','automobile','bird','cat','deer', 'dog
model = models.Sequential()
# SOLUTION: Add an explicit Input layer
model.add(layers.Input(shape=(32, 32, 3)))
# Now define the rest of the model without 'input_shape' in the
model.add(layers.Conv2D(32, (3,3), activation='relu'))
model.add(layers.MaxPooling2D((2,2)))
model.add(layers.Conv2D(64, (3,3), activation='relu'))
model.add(layers.MaxPooling2D((2,2)))
model.add(layers.Conv2D(64, (3,3), activation='relu'))
model.add(layers.Flatten())
model.add(layers.Dense(64, activation='relu'))
model.add(layers.Dense(10, activation='softmax'))
# Compile and Fit
model.compile(optimizer='adam',
loss='sparse_categorical_crossentropy',
metrics=['accuracy'])
model.fit(x_train, y_train, epochs=10,validation_data=(x_test, y
```

```
Epoch 1/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 70s 44ms/step - accuracy: 0.3577 - loss: 1.7412 - val_accuracy: 0.5486 - val_loss: 1.2501
Epoch 2/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 65s 41ms/step - accuracy: 0.5881 - loss: 1.1575 - val_accuracy: 0.6297 - val_loss: 1.0495
Epoch 3/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 83s 42ms/step - accuracy: 0.6526 - loss: 0.9920 - val_accuracy: 0.6431 - val_loss: 1.0035
Epoch 4/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 81s 41ms/step - accuracy: 0.6879 - loss: 0.8886 - val_accuracy: 0.6785 - val_loss: 0.9231
Epoch 5/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 64s 41ms/step - accuracy: 0.7223 - loss: 0.7999 - val_accuracy: 0.6991 - val_loss: 0.8709
Epoch 6/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 66s 42ms/step - accuracy: 0.7424 - loss: 0.7373 - val_accuracy: 0.6990 - val_loss: 0.8677
Epoch 7/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 64s 41ms/step - accuracy: 0.7612 - loss: 0.6824 - val_accuracy: 0.7073 - val_loss: 0.8749
Epoch 8/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 82s 41ms/step - accuracy: 0.7801 - loss: 0.6313 - val_accuracy: 0.7177 - val_loss: 0.8492
Epoch 9/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 65s 42ms/step - accuracy: 0.7939 - loss: 0.5855 - val_accuracy: 0.7203 - val_loss: 0.8439
Epoch 10/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 87s 44ms/step - accuracy: 0.8099 - loss: 0.5420 - val_accuracy: 0.7171 - val_loss: 0.8539
<keras.src.callbacks.history.History at 0x7d82f543f110>
```

```
index = 1010
input_image = x_test[index]
input_label = y_test[index][0]
image_input = np.expand_dims(input_image, axis=0)
prediction = model.predict(image_input)
predicted_class = class_names[np.argmax(prediction)]
actual_class = class_names[input_label]
plt.imshow(input_image)
plt.title(f"Actual: {actual_class} | Predicted: {predicted_class
plt.axis("off")
plt.show()
```

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1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 124ms/step
```

Actual: airplane | Predicted: airplane